

UNIQUENESS OF WEIGHTS FOR RECURRENT NETS

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1. Introduction

We study recurrent neural networks evolving either in discrete or in continuous time. The dynamics of such systems can be described by a set of either difference or differential equations of the following type (to simplify notations, we use the superscripts “+” and “.” to denote time-shift and time-derivative respectively, and we omit the time arguments t):

$$\begin{aligned} x^+ \text{ (or } \dot{x} \text{)} &= \vec{\sigma}(Ax + Bu) \\ y &= Cx, \end{aligned} \tag{1}$$

with $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, and where $\vec{\sigma}(x) = (\sigma(x_1), \dots, \sigma(x_n))$, with σ a function from $\mathbb{R}$ to itself. (For the continuous-time case, we always assume that the function σ is globally Lipschitz, so that the existence and uniqueness of solutions for the differential equation is guaranteed.)

We will consider the *parameter identifiability* problem, which asks about the possibility of recovering the entries of the matrices A , B , and C from the input/output map $u(\cdot) \mapsto y(\cdot)$ of the system. This question has been already addressed in [1] and in [2] (and, for feedforward nets, in [7] and in [5]) where it is proved that, under appropriate minimality assumptions, the *zero-initial state* i/o behavior determines, up to a small number of symmetries, the weights of the model. In this paper we prove the same result for *arbitrary-initial state* i/o maps. Moreover, we show that, for a generic subclass of these models, the minimality conditions needed in order for the results to hold are exactly the observability conditions found in the recent paper [3]. It is interesting to notice that, inside this subclass, these observability conditions are also necessary for identifiability.

2. The activation function σ

Given any map $\sigma : \mathbb{R} \rightarrow \mathbb{R}$, we say that σ satisfies the *independence property* (“IP” from now on) if, for every positive integer l , nonzero real numbers $b_1, \dots, b_l$, and real numbers $\beta_1, \dots, \beta_l$ such that $(b_i, \beta_i) \neq \pm(b_j, \beta_j) \ \forall i \neq j$, it must hold that the functions $1, \sigma(b_1x + \beta_1), \dots, \sigma(b_lx + \beta_l)$ are linearly independent. The function $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ which

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appears in our model will always assumed to satisfy property **IP** and to be odd. The next result, which was proved in [4], Theorem 1, gives a condition for the **IP** property, weak enough to include many of the functions used for neural network applications:

*Assume that σ is a real-analytic function, and it extends to an analytic function $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ defined on a subset $D \subseteq \mathbb{C}$ of the form $D = \{z \mid |\operatorname{Im} z| \leq \lambda\} \setminus \{z_0, \bar{z}_0\}$ for some $\lambda > 0$, where $\operatorname{Im} z_0 = \lambda$ and z_0 and $\bar{z}_0$ are singularities, that is, there is a sequence $z_n \rightarrow z_0$ so that $|\sigma(z_n)| \rightarrow \infty$, and similarly for $\bar{z}_0$. Then, σ satisfies property **IP**.*

Given a matrix M , we denote by M_i the i -th row of M . For any two positive integer n, m , we let:

$$\mathcal{B}_{n,m} = \left\{ B \in \mathbb{R}^{n \times m} \mid \begin{array}{l} B_i \neq 0 \quad \forall i = 1, \dots, n \\ B_i \neq \pm B_j \quad \forall i \neq j \end{array} \right\}.$$

The next two Lemmas follow from the definition of the property **IP**, and from the fact that if $B \in \mathcal{B}_{n,m}$, then there exists $u \in \mathbb{R}^m$ such that $B_i u \neq 0$ for all i and $B_i u \neq \pm B_j u$ for all $i \neq j$.

Lemma 2.1 Let $B \in \mathcal{B}_{n,m}$. If $\sum_{i=1}^n c_i \sigma(\alpha_i + B_i u) = 0$ for all $u \in \mathbb{R}^m$, then $c_i = 0$ for all $i = 1, \dots, n$.

If $v, w \in \mathbb{R}^m$, we denote by $\langle v, w \rangle$ their scalar product.

Lemma 2.2 Let $B^1 \in \mathcal{B}_{n_1,m}$, $B^2 \in \mathcal{B}_{n_2,m}$, and $\gamma_i^j \in \mathbb{R}$ with $j = 1, 2$ and $1 \leq i \leq n_j$. If, for some all nonzero coefficients α_j^i we have:

$$\sum_{j=1}^{n_1} \alpha_j^1 \sigma(\gamma_j^1 + \langle B_j^1, u \rangle) = \sum_{j=1}^{n_2} \alpha_j^2 \sigma(\gamma_j^2 + \langle B_j^2, u \rangle)$$

for all $u \in \mathbb{R}^m$, then we can conclude that: (1) $n_1 = n_2 = n$, and (2) there exist a permutation π of the set $\{1, \dots, n\}$, and some coefficients $\beta(i) = \pm 1$, for $i = 1, \dots, n$, such that:

$$\gamma_j^1 = \beta(j) \gamma_{\pi(j)}^2, \quad B_j^1 = \beta(j) B_{\pi(j)}^2, \quad \alpha_j^1 = \beta(j) \alpha_{\pi(j)}^2.$$

3. Statement of the main results

Assume we are given a σ -system $\Sigma \equiv (A, B, C)_\sigma$ (either in discrete or in continuous time) initialized at a given state x_0 (we will write $\Sigma \equiv (A, B, C, x_0)_\sigma$ to denote the σ -system Σ together with the initial state x_0). Then we can associate to Σ an i/o map as follows. In the discrete-time case, for any sequence of inputs $u_1, \dots, u_k$ a sequence of outputs $y_0 = Cx_0, y_1, \dots, y_k$ is generated. In the continuous-time case for any control function $u(\cdot) : [0, T] \rightarrow \mathbb{R}^m$ (which we assume to be measurable essentially bounded), after solving the differential equation with initial state x_0 and denoting its solution with $x(t)$, we have a corresponding output function $y(t) = Cx(t)$. Thus, in both cases, to the given initialized system $\Sigma \equiv (A, B, C, x_0)_\sigma$ we associate an i/o map:

$$\lambda_{\Sigma, x_0} : \begin{cases} (u_1, \dots, u_k) & \rightarrow (y_0, \dots, y_k) & \text{discrete} \\ u(\cdot) & \rightarrow y(\cdot) & \text{continuous.} \end{cases}$$

We say that two initialized σ -system $\Sigma \equiv (A, B, C, x_0)_\sigma$, and $\tilde{\Sigma} \equiv (\tilde{A}, \tilde{B}, \tilde{C}, \tilde{x}_0)_\sigma$ are *i/o equivalent* if $p = \tilde{p}$, $m = \tilde{m}$, and $\lambda_{\Sigma, x_0} = \lambda_{\tilde{\Sigma}, \tilde{x}_0}$. We will study the problem of determining when two given initialized σ -systems are i/o equivalent.

Let:

$$\Lambda^n = \left\{ T \in \text{Gl}(n) \mid T = PQ \text{ with } \begin{array}{l} P = \text{permutation matrix} \\ Q = \text{Diag}(\beta(1), \dots, \beta(n)), \beta(i) = \pm 1. \end{array} \right\}.$$

Definition 3.3 We say that two initialized σ -systems $\Sigma_i = (D_i, A_i, B_i, C_i, x_i)_\sigma$, for $i = 1, 2$ are (sign-permutation) *equivalent* if $n_1 = n_2 = n$, and if there exists a matrix $T \in \Lambda^n$ such that:

$$A_2 = T^{-1}A_1T, \quad C_2 = C_1T, \quad B_2 = T^{-1}B_1, \quad x_2 = T^{-1}x_1.$$

Since σ was assumed to be odd, it is easy to see that two equivalent σ -systems are i/o equivalent. Our aim is to show that *generically* also the converse holds. We say that a subset of $\mathbb{R}^p$ is *generic* if it is nonempty and if its complement is the set of zeroes of a finite number of polynomials in p variables. By a generic subset of σ -systems we mean a generic subset of $\mathbb{R}^{n^2+nm+np}$ when we identify the set of all σ -systems with $\mathbb{R}^{n^2+nm+np}$.

We first let $\mathcal{S}$ to be the subset of σ -systems (1) for which the matrix B is in $\mathcal{B}_{n,m}$. Notice that $\mathcal{S}$ is a generic subset of σ -systems.

We say that a subspace V of $\mathbb{R}^n$ is a *coordinate subspace* if it is of the type:

$$V = \text{span} \{ e_{i_1}, \dots, e_{i_k} \},$$

where e_{i_j} are the vectors of the canonical basis in $\mathbb{R}^n$. For any pair of matrices (A, C) we denote by $\mathcal{O}_c(A, C)$ the largest A -invariant coordinate subspace included in $\ker C$ (see Corollary 4.8 for a way to compute $\mathcal{O}_c(A, C)$). A system Σ is said to be *observable* if for each two distinct initial states there exists some control that gives different output when Σ is started at those states (for a precise statement, both for discrete and continuous time systems, see e.g. [6]). Given these definitions, the following result holds (see [3], Theorem 1):

Proposition 3.4 A system $\Sigma \in \mathcal{S}$ is observable if and only if $\ker A \cap \ker C = \mathcal{O}_c(A, C) = 0$.

Now, we can state our main results, which will be proved in a later section.

Theorem 1 Assume that σ is an odd map which satisfies property **IP**, (and, for the continuous-time case, it is globally Lipschitz). Let $\Sigma \equiv (A, B, C, x_0)_\sigma$ and $\tilde{\Sigma} \equiv (\tilde{A}, \tilde{B}, \tilde{C}, \tilde{x}_0)_\sigma$ be two observable σ -systems in $\mathcal{S}$. Then, these systems are i/o equivalent if and only if they are equivalent.

Remark 3.5 Notice that, inside the class $\mathcal{S}$, the observability condition is also necessary in order to guarantee the implication

$$\text{i/o equivalence} \quad \Rightarrow \quad \text{equivalence}.$$

More precisely, if $\mathcal{S}_0 \subseteq \mathcal{S}$ is any class of systems so that this implication holds for any pair of systems in $\mathcal{S}_0$ and initial states, then every system in $\mathcal{S}_0$ must be observable. This is proved as follows. Assume that $\Sigma \equiv (A, B, C)_\sigma$ is not observable. By definition of observability, there exist then two *distinct* states x_1 and x_2 which are not distinguishable. This means precisely that $\Sigma_1 \equiv (A, B, C, x_1)_\sigma$ and $\Sigma_2 \equiv (A, B, C, x_2)_\sigma$ are i/o equivalent. If the above implication would hold, then the systems must be equivalent. So there is a $T \in \Lambda^n$ such that $TB = B$ and $Tx_2 = x_1$. But the fact that $B \in \mathcal{B}_{n,m}$ implies that T is the identity. (In other words, the action of Λ^n on $\mathcal{S}$ is a free group action.) Thus $x_1 = x_2$, a contradiction.

Remark 3.6 It is also interesting to notice that, if a σ -system is not observable because $\mathcal{O}_c(A, C) \neq 0$, a reduction by unobservability is possible. It is only necessary to note that one can induce a dynamics on the quotient space $\mathbb{R}^n / \mathcal{O}_c(A, C)$, which can be done because the subspace $\mathcal{O}_c(A, C)$ is an A -invariant coordinate subspace. More precisely, the construction is as follows.

Let $\Sigma \in \mathcal{S}$. If $\mathcal{O}_c(A, C) \neq 0$, then after if necessary reordering variables, the equations for Σ take the following block form. The first block of variables corresponds to a basis of $\mathcal{O}_c(A, C)$, and has size $n_1 = \text{dimension of } \mathcal{O}_c(A, C)$; the second set of variables has size $n_2 = n - n_1$.

$$\begin{aligned} x_1^+ &= \sigma(A_1 x_1 + A_2 x_2 + B_1 u) \\ x_2^+ &= \sigma(A_3 x_2 + B_2 u) \\ y &= C_2 x_2. \end{aligned}$$

Then $(A, B, C, (x_1, x_2))_\sigma$ and $\Sigma' \equiv (A_3, B_2, C_2, x_2)_\sigma$ are i/o-equivalent, for any initial state $x = (x_1, x_2)$ of the original system, and the second system has lower dimension. The system Σ' is again in $\mathcal{S}$. Moreover, Σ' is observable if $\ker A \cap \ker C = 0$. Indeed, the choice of variables insures that $\mathcal{O}_c(A_3, C_2) = 0$. Furthermore,

$$\text{rank} \begin{pmatrix} A_1 & A_2 \\ 0 & A_3 \\ 0 & C_2 \end{pmatrix} = n$$

implies

$$\text{rank} \begin{pmatrix} 0 & A_3 \\ 0 & C_2 \end{pmatrix} = n_2$$

so the conclusion $\ker A_3 \cap \ker C_2 = 0$ follows from $\ker A \cap \ker C = 0$.

4. Some Technical Properties

In this section we prove some technical facts which will be used later. (Notice that both the statements of the results and the proofs are the same for the discrete and the continuous time case.) For any σ -system Σ , we let:

$$\begin{aligned} I_0(\Sigma) &= \{i \mid \exists j \in \{1, \dots, p\} \text{ and } c_{ji} \neq 0\}, \\ I_k(\Sigma) &= \{i \mid \exists l \in I_{k-1}(\Sigma) \text{ with } a_{li} \neq 0\}, \end{aligned}$$

and

$$I(\Sigma) = \bigcup_{k \geq 0} I_k(\Sigma).$$

We also denote

$$I^c(\Sigma) = \{1, \dots, n\} \setminus I(\Sigma).$$

We use the following notation. For each subset $J \subseteq \{1, \dots, n\}$, let V_J be the following coordinate subspace:

$$V_J = \text{span} \{ e_j \mid j \in J \}.$$

Lemma 4.7 Let $\Sigma \equiv (A, B, C)_\sigma$ be a σ -system, and let J be any subset as above. Then,

$$V_J \text{ is } A\text{-invariant and included in } \ker C \Rightarrow J \subseteq I^c(\Sigma).$$

Moreover,

$$V_{I^c(\Sigma)} \text{ is } A\text{-invariant and included in } \ker C.$$

Proof. Assume first that V_J is A -invariant and is contained in $\ker C$. Suppose that $J \cap I(\Sigma) \neq \emptyset$. Thus $J \cap I_k(\Sigma) \neq \emptyset$ for some k , and we may assume that k is minimal with this property. Let $i \in J \cap I_k(\Sigma)$. If $k = 0$ then $Ce_i \neq 0$, contradicting $V_J \subseteq \ker C$. Thus there is some $l \in I_{k-1}(\Sigma)$ with $a_{li} \neq 0$. Since V_J is A -invariant, this means that $e_l \in V_J$, that is, l is also in J . But this contradicts the minimality of k .

Let now $J = I^c(\Sigma)$, and write $I = I(\Sigma)$. Since $I_0(\Sigma) \subseteq I$, it is clear that $V_J \subseteq \ker C$, thus we need only to show that V_J is A -invariant. Notice that A -invariance is equivalent to: $a_{lj} = 0$ for all $l \notin J$ and all $j \in J$. Fix any $l \notin J$. Then $l \in I$, and so there exists k such that $l \in I_k(\Sigma)$. By definition $I_{k+1}(\Sigma)$ contains all the indices j such that $a_{lj} \neq 0$. Thus for each $i \in J$, $a_{li} = 0$, as desired. $\blacksquare$

Since $\mathcal{O}_c(A, C)$ is by definition the largest A -invariant coordinate subspace included in $\ker C$, we conclude:

Corollary 4.8 $\mathcal{O}_c(A, C) = V_{I^c(\Sigma)}$.

Note that this observation provides an easy algorithm for obtaining $\mathcal{O}_c(A, C)$. Also, it implies that $I^c(\Sigma)$ must be empty if $\mathcal{O}_c(A, C) = 0$; we state this for further reference:

Corollary 4.9 If $\mathcal{O}_c(A, C) = 0$, then $I(\Sigma) = \{1, \dots, n\}$.

5. Proof of Theorem 1

From now on, we assume that we are given two i/o equivalent σ -systems Σ and $\tilde{\Sigma}$ in $\mathcal{S}$, initialized at x_0 and $\tilde{x}_0$, and with state space dimensions n and $\tilde{n}$ respectively.

First we introduce some useful notations. In the case of discrete-time σ -systems, for any sequence of control values $u_1, \dots, u_k \in \mathbb{R}^m$, we let, for $i = 1, \dots, k$:

$$\begin{aligned} x_i[u_1, \dots, u_i] &= \sigma(Ax_{i-1}[u_1, \dots, u_{i-1}] + Bu_i), \\ \tilde{x}_i[u_1, \dots, u_i] &= \sigma(\tilde{A}\tilde{x}_{i-1}[u_1, \dots, u_{i-1}] + \tilde{B}u_i), \end{aligned}$$

and, for $i = 0, \dots, k$:

$$y_i[u_1, \dots, u_i] = Cx_i[u_1, \dots, u_i], \quad \tilde{y}_i[u_1, \dots, u_i] = \tilde{C}\tilde{x}_i[u_1, \dots, u_i].$$

Notice that since Σ and $\tilde{\Sigma}$ are i/o equivalent, we have in particular that for any sequence $u_1, \dots, u_r, u \in \mathbb{R}^m$:

$$y_{r+1}[u_1, \dots, u_r, u] = \tilde{y}_{r+1}[u_1, \dots, u_r, u].$$

The previous equation can be rewritten in coordinate form as:

$$\sum_{\{i|c_{ji} \neq 0\}} c_{ji} \sigma((Ax_r[u_1, \dots, u_r])_i + B_i u) = \sum_{\{i|\tilde{c}_{ji} \neq 0\}} \tilde{c}_{ji} \sigma((\tilde{A}\tilde{x}_r[u_1, \dots, u_r])_i + \tilde{B}_i u), \quad (2)$$

for each $j = 1, \dots, p$.

Consider now the case of continuous-time σ -systems. Our aim next will be to show that also in this case an equation similar to (2) holds. For any control function $w(\cdot) : [0, T_w] \rightarrow \mathbb{R}^m$, with $T_w \geq 0$, we denote by $x_w(t)$ and by $\tilde{x}_w(t)$ the corresponding trajectories in Σ and $\tilde{\Sigma}$, starting at x_0 and $\tilde{x}_0$ respectively. Moreover, we let $y_w(t) = Cx_w(t)$ and $\tilde{y}_w(t) = \tilde{C}\tilde{x}_w(t)$ be the output signals generated.

Since Σ and $\tilde{\Sigma}$ are i/o equivalent, we have that for all $w(\cdot) : [0, T_w] \rightarrow \mathbb{R}^m$,

$$y_w(t) = \tilde{y}_w(t)$$

for all $t \in [0, T_w]$. If the control w happens to be continuous as a function of time, then the solutions of the corresponding differential equations are differentiable (as opposed to merely absolutely continuous) functions. Thus, in that case, also, for each $\bar{t} \in [0, T_w]$:

$$C\dot{x}_w(t)|_{t=\bar{t}} = \dot{y}_w(t)|_{t=\bar{t}} = \dot{\tilde{y}}_w(t)|_{t=\bar{t}} = \tilde{C}\dot{\tilde{x}}_w(t)|_{t=\bar{t}}. \quad (3)$$

Now take any control function $w(\cdot) : [0, T_w] \rightarrow \mathbb{R}^m$, with $T_w \geq 0$. Since the two initialized systems Σ and $\tilde{\Sigma}$ are i/o equivalent, also the systems $\Sigma \equiv (A, B, C, x_w(T_w))_\sigma$ and $\tilde{\Sigma} \equiv (\tilde{A}, \tilde{B}, \tilde{C}, \tilde{x}_w(T_w))_\sigma$ are i/o equivalent. Applying now the constant control $w \equiv u$, for any value $u \in \mathbb{R}^m$, and taking $\bar{t} = 0$, from (3) there results:

$$C\vec{\sigma}(Ax_w(T_w) + Bu) = \tilde{C}\vec{\sigma}(\tilde{A}\tilde{x}_w(T_w) + \tilde{B}u). \quad (4)$$

Now we prove a preliminary result.

Proposition 5.10 Let $\Sigma \equiv (A, B, C, x_0)_\sigma$ and $\tilde{\Sigma} \equiv (\tilde{A}, \tilde{B}, \tilde{C}, \tilde{x}_0)_\sigma$ be two i/o equivalent σ -systems. If $\mathcal{O}_c(A, C) = \mathcal{O}_c(\tilde{A}, \tilde{C}) = 0$, then $n = \tilde{n}$ and there exists a matrix $T \in \Lambda^n$ such that, $\tilde{C} = CT$, and $\tilde{B} = T^{-1}B$.

Moreover, for the discrete-time case, for each $r \geq 0$ and each sequence $u_1, \dots, u_r$, we have that:

$$\tilde{A}\tilde{x}_r[u_1, \dots, u_r] = T^{-1}Ax_r[u_1, \dots, u_r]. \quad (5)$$

For the continuous-time case, for each $t \geq 0$, and each control function $w(\cdot) : [0, t] \rightarrow \mathbb{R}^m$, we have that:

$$\tilde{A}\tilde{x}_w(t) = T^{-1}Ax_w(t). \quad (6)$$

Proof. We deal with both cases at the same time. First we want to show that for each $i \in \{1, \dots, n\}$ there exist $\pi(i) \in \{1, \dots, \tilde{n}\}$ and $\beta(i) = \pm 1$ such that:

$$B_i = \beta(i)\tilde{B}_{\pi(i)}, \quad (7)$$

and, for the discrete-time case, for any $r \geq 0$, and any choice of $u_i \in \mathbb{R}^m$:

$$(Ax_r[u_1, \dots, u_r])_i = \beta(i)(\tilde{A}\tilde{x}_r[u_1, \dots, u_r])_{\pi(i)}, \quad (8)$$

for the continuous-time case, for any $t \geq 0$ and any control function $w(\cdot) : [0, t] \rightarrow \mathbb{R}^m$:

$$(Ax_w(t))_i = \beta(i)(\tilde{A}\tilde{x}_w(t))_{\pi(i)}. \quad (9)$$

Notice that for any fixed $1 \leq i \leq n$, the index $\pi(i) \in \{1, \dots, \tilde{n}\}$ satisfying the previous equations, if it exists, is unique. In fact, if we assume that there exist two indices $\pi(i)$ and $\pi'(i)$ for which equation (7) holds, then we get that $\tilde{B}_{\pi(i)} = \pm \tilde{B}_{\pi'(i)}$, which contradicts the assumption that $\tilde{B} \in \mathcal{B}_{\tilde{n}, m}$. Similarly, the $\beta(i)$ are unique. Moreover, if we prove that (7) and (8) or (9) hold, we get that necessarily $n \leq \tilde{n}$, thus by symmetry we also have that $\tilde{n} \leq n$; so we can conclude that $n = \tilde{n}$.

Fix any $\bar{i} \in \{1, \dots, n\}$. Then by Corollary 4.9 there exists a $k \geq 0$ such that $\bar{i} \in I_k(\Sigma)$. We will construct $\pi(\bar{i})$ and $\beta(\bar{i})$ and prove that equations (7) and (8) or (9) hold by induction on the minimal such k .

Assume that $k = 0$. Then, by definition of $I_0(\Sigma)$, there exists some $1 \leq \bar{j} \leq p$ such that $c_{\bar{j}\bar{i}} \neq 0$. For the discrete-time case, i/o equivalence implies that equation (2) holds. Thus, in particular, we have:

$$\sum_{\{i|c_{\bar{j}i} \neq 0\}} c_{\bar{j}i} \sigma((Ax_r[u_1, \dots, u_r])_i + B_i u) = \sum_{\{i|\tilde{c}_{\bar{j}i} \neq 0\}} \tilde{c}_{\bar{j}i} \sigma((\tilde{A}\tilde{x}_r[u_1, \dots, u_r])_i + \tilde{B}_i u), \quad (10)$$

for any sequence $u_1, \dots, u_r, u \in \mathbb{R}^m$. For the continuous-time case, i/o equivalence implies that equation (4) holds. So we also have:

$$\sum_{\{i|c_{\bar{j}i} \neq 0\}} c_{\bar{j}i} \sigma((Ax_w(t))_i + B_i u) = \sum_{\{i|\tilde{c}_{\bar{j}i} \neq 0\}} \tilde{c}_{\bar{j}i} \sigma((\tilde{A}\tilde{x}_w(t))_i + \tilde{B}_i u), \quad (11)$$

for any $t \geq 0$ and any control function $w(\cdot) : [0, t] \rightarrow \mathbb{R}^m$. Notice that both equations (10) and (11) hold for all $u \in \mathbb{R}^m$. Moreover $x_r[u_1, \dots, u_r]$ and $\tilde{x}_r[u_1, \dots, u_r]$, and also $x_w(t)$ and $\tilde{x}_w(t)$ do not depend on u . Thus, we can apply Lemma 2.2, which gives us the desired $\pi(\bar{i})$ and $\beta(\bar{i})$. (Note that, by the uniqueness property remarked above, these $\pi(\bar{i})$ and $\beta(\bar{i})$ do not depend on the particular inputs used in the argument.)

Assume that now $k > 0$. Then, by definition of $I_k(\Sigma)$ there exists $\bar{j} \in I_{k-1}(\Sigma)$ such that $a_{\bar{j}\bar{i}} \neq 0$. By inductive assumption we have that there exists $\pi(\bar{j})$ and $\beta(\bar{j})$ such that:

$$\begin{cases} (Ax_{r+1}[u_1, \dots, u_r, u_{r+1}])_{\bar{j}} = \beta(\bar{j})(\tilde{A}\tilde{x}_{r+1}[u_1, \dots, u_r, u_{r+1}])_{\pi(\bar{j})}, & \text{discrete-time} \\ (Ax_w(t))_{\bar{j}} = \beta(\bar{j})(\tilde{A}\tilde{x}_w(t))_{\pi(\bar{j})}, & \text{continuous-time} \end{cases} \quad (12)$$

For the discrete-time case, we have just to rewrite the first of the previous equations as:

$$\begin{aligned} \sum_{\{i|a_{\bar{j}i} \neq 0\}} a_{\bar{j}i} \sigma((Ax_r[u_1, \dots, u_r])_i + B_i u_{r+1}) = \\ \sum_{\{i|\tilde{a}_{\bar{j}i} \neq 0\}} \beta(\bar{j}) \tilde{a}_{\bar{j}i} \sigma((\tilde{A}\tilde{x}_r[u_1, \dots, u_r])_i + \tilde{B}_i u_{r+1}), \end{aligned}$$

and, applying again Lemma 2.2, since $a_{\bar{j}i} \neq 0$, we obtain $\pi(\bar{i})$ and $\beta(\bar{i})$, as desired.

For the continuous-time case, pick any $\bar{t} \geq 0$, and any control function $w(\cdot) : [0, \bar{t}] \rightarrow \mathbb{R}^m$. Fix $\epsilon > 0$, and, for each $u \in \mathbb{R}^m$, consider the following control function:

$$w^u(t) = \begin{cases} w(t) & t \in [0, \bar{t}] \\ u & t \in [\bar{t}, \bar{t} + \epsilon]. \end{cases}$$

Then we have $x_w(\bar{t}) = x_{w^u}(\bar{t})$ (same for $\tilde{x}_w(\bar{t}) = \tilde{x}_{w^u}(\bar{t})$). Moreover since w^u is constant (and hence continuous) for all $t > \bar{t}$, $x_{w^u}(\cdot)$ and $\tilde{x}_{w^u}(\cdot)$ are differentiable for all $t > \bar{t}$. Thus differentiating the second equation in (12), and then taking the limit as $t \rightarrow \bar{t}$, we have:

$$\sum_{\{i|a_{\bar{j}i} \neq 0\}} a_{\bar{j}i} \sigma((Ax_w(\bar{t}))_i + B_i u) = \sum_{\{i|\tilde{a}_{\bar{j}i} \neq 0\}} \beta(\bar{j}) \tilde{a}_{\bar{j}i} \sigma((\tilde{A}\tilde{x}_w(\bar{t}))_i + \tilde{B}_i u).$$

Now we apply again Lemma 2.2, to get $\pi(\bar{i})$ and $\beta(\bar{i})$, as desired.

Thus equations (7) and (8) or (9) hold. So, in particular we know that $n = \tilde{n}$.

Using now equations (7), the fact that σ is an odd function, and equations (8) or (9), we can rewrite the equation $y_1(u) = \tilde{y}_1(u)$, for the discrete-time case, and the equation $\dot{y}(t)|_{t=0} = \dot{\tilde{y}}(t)|_{t=0}$ for the continuous-time case (using a constant, hence continuous, control u), as:

$$\sum_{i=1}^n (c_{ij} - \beta(i) \tilde{c}_{j\pi(i)}) \sigma((Ax_0)_i + B_i u) = 0.$$

from the previous equation together with Lemma 2.1, we get:

$$c_{ji} = \tilde{c}_{j\pi(i)}. \quad (13)$$

Let now T^{-1} be the matrix PQ , where P is the permutation matrix representing $\pi(i)$, i.e. $Pe_i = e_{\pi(i)}$, and $Q = \text{Diag}(\beta(1), \dots, \beta(n))$. Then, it is easy to verify that equations (7), (8) or (9), and (13) say that:

$$\tilde{C} = CT, \quad \tilde{B} = T^{-1}B,$$

and that equations (5) or (6) hold. Thus, since $T \in \Lambda^n$, our statement is proved. $\blacksquare$

Now we can prove Theorem 1.

Proof of Theorem 1 We need only to prove that i/o equivalence implies equivalence. from Proposition 5.10, we already know that $n = \tilde{n}$ and there exists a matrix $T \in \Lambda^n$ which satisfies $\tilde{C} = CT$, $\tilde{B} = T^{-1}B$, and also equations (5) or (6).

For discrete-time σ -systems, we can rewrite equation (5), with $r = 1$, to get:

$$\sum_{j=1}^n a_{ij} \sigma((Ax_0)_j + B_j u) = \beta(i) \sum_{j=1}^n \tilde{a}_{\pi(i)j} \sigma((\tilde{A}\tilde{x}_0)_j + \tilde{B}_j u), \quad \forall 1 \leq i \leq n \quad (14)$$

for all u . Using the facts that $(\tilde{A}\tilde{x}_0)_{\pi(j)} = \beta(j)(Ax_0)_j$ (case $r = 0$ in the Proposition) and $\tilde{B}_{\pi(j)} = \beta(j)B_j$, together with the fact that σ is odd, the previous equation becomes:

$$\sum_{j=1}^n (a_{ij} - \beta(i)\tilde{a}_{\pi(i)\pi(j)})\sigma((Ax_0)_j + B_j u) = 0.$$

By applying Lemma 2.1 to the previous equation we conclude:

$$a_{ij} = \beta(i)\tilde{a}_{\pi(i)\pi(j)},$$

which means that

$$\tilde{A} = T^{-1}AT.$$

To get the similar conclusion for the continuous-time case, we have just to differentiate equation (6) (again, applied to a constant control) with respect to t and evaluate at $t = 0$. This will give us again equation (14), from which we conclude as before.

Now it remains only to show that $\tilde{x}_0 = T^{-1}x_0$. To get this equation, we argue as follows. First we get:

$$Cx_0 = \tilde{C}\tilde{x}_0 = CT\tilde{x}_0,$$

Moreover, we also have:

$$Ax_0 = T\tilde{A}\tilde{x}_0 = AT\tilde{x}_0.$$

Thus $x_0 - T\tilde{x}_0 \in \ker A \cap \ker C$. By observability we obtain that $x_0 = T\tilde{x}_0$, as desired.

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